

FYS4480/9480 September 24



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variations of coefficients
and HF-theory

$$|p\rangle = \sum_{\lambda} c_{p\lambda} |\lambda\rangle$$

$$\hat{h}_0 |\lambda\rangle = \epsilon_{\lambda} |\lambda\rangle$$

$$\hat{h}_0 = \hat{t} + \hat{u}_{ext}$$

$$\langle p | q \rangle = \delta_{pq}$$

$$\langle \lambda | \sigma \rangle = \delta_{\lambda\sigma}$$



$$|\underline{\Phi}_0\rangle = \frac{1}{\sqrt{n!}} \begin{vmatrix} \varphi_0(1) & \dots & \varphi_0(n) \\ \varphi_1(1) & & \vdots \\ \vdots & & \vdots \\ \varphi_n(1) & & \varphi_n(n) \end{vmatrix}$$

$$\begin{aligned} \langle \underline{\Phi}_0 | H | \underline{\Phi}_0 \rangle &= \sum_{\lambda \in F} \underbrace{\langle \lambda | \vec{h}_0 | \lambda \rangle}_{\varepsilon_\lambda} \\ &+ \frac{1}{2} \sum_{\lambda \notin F} \langle \lambda \pm | \vec{v}^2 | \lambda \pm \rangle_{AS} \end{aligned}$$

$$|\underline{\Phi}_0^{HF}\rangle = \frac{1}{\sqrt{n!}} \begin{vmatrix} \varphi_0(1) & \dots & \varphi_0(n) \\ \varphi_1(1) & & \vdots \\ \vdots & & \vdots \\ \varphi_n(1) & & \varphi_n(n) \end{vmatrix}$$

$$\det(\Phi_0^{HF}) = \det(C) \det(\Phi)$$

$$\langle \Phi_0^{HF} | H | \Phi_0^{HF} \rangle =$$

$$\sum_{i' \leq F} \langle i' | h_0 | i' \rangle + \frac{1}{2} \sum_{i' j' \leq F} \langle i' j' | v | i' j' \rangle_A$$

insert $|p\rangle = \sum_{\lambda} \langle \phi_{\lambda} | \lambda \rangle$

$$E[\Phi_0^{HF}] = \sum_{i'=1}^F \sum_{\alpha \beta} C_{i'\alpha}^* C_{i'\beta} \langle \alpha | h_0 | \beta \rangle$$

$$+ \frac{1}{2} \sum_{i' j'=1}^N \sum_{\alpha \beta \gamma \delta} C_{i'\alpha}^* C_{j'\beta}^* C_{i'\gamma} C_{j'\delta} \langle \alpha \beta | v | \gamma \delta \rangle_A$$

Functional

$$F[\Phi_c^{HF}] = E[\Phi_c^{HF}]$$

$$= \sum_{i=1}^N \epsilon_i \sum_{\alpha} |c_{i\alpha}|^2$$

decouple $c_{i\alpha}^*$ and $c_{i\alpha}$

Differentiate wrt $c_{i\alpha}^*$

$$\frac{\delta F[\Phi_c^{HF}]}{\delta c_{i\alpha}^*} = 0 \quad ; \quad \text{pick one } c_{i\alpha}^*$$

$i\alpha$ - fixed

$$\sum_{\beta} c_{i\beta} \langle \alpha | \hat{h}_0 | \beta \rangle +$$

$$\sum_{j=1}^N \sum_{\beta \delta} c_{j\beta}^* c_{j\delta} c_{i\gamma} \langle \alpha \beta | \hat{v} | \gamma \delta \rangle_{AS} = \epsilon_i \epsilon_{i\alpha}$$

relabel $\beta \leftrightarrow \gamma$

$$\sum_{\beta} \left\{ \langle \alpha | \hat{h}_0 | \beta \rangle + \sum_{j=1}^N \sum_{\gamma \delta} c_{j\gamma}^* c_{j\delta} \times \langle \alpha \gamma | \hat{v} | \beta \delta \rangle_{AS} \right\} c_{i\beta} = \epsilon_i \epsilon_{i\alpha}$$

Define

$$h_{\alpha\beta}^{\text{HF}} = \langle \alpha | \hat{h}_c | \beta \rangle + \sum_{j=1}^N \sum_{\gamma\delta} C_j^* C_j \delta \times \langle \alpha\gamma | r | \beta\delta \rangle_{\text{AV}}$$

$$\sum_P h_{\alpha\beta}^{\text{HF}} C_{i\beta} = \epsilon_i^{\text{HF}} C_{i\alpha}$$

$\vec{C}$ which contains all $C_{i\alpha}$
 $\hat{h}^{\text{HF}} \vec{C} = \epsilon \vec{C}$

Standard HF-recipe:

- solve by iteration

- start with a guess $C_{i\beta}^{(0)}$

$$C^{(0)} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 0 & \\ & & & 1 \end{bmatrix}$$

- solve $h^{HF}(0)$

$$h_{\alpha\beta}^{HF(0)} = \langle \alpha | h_0 | \beta \rangle + \sum_{j=1}^N \sum_{\delta} C_{j\delta}^{*(0)} C_{j\delta}^{(0)} \langle \alpha\delta | r | \beta\delta \rangle$$

- Diagonalize

$$H^{HF}(0) \vec{C}^{(1)} = E^{HF}(0) \vec{C}^{(1)}$$

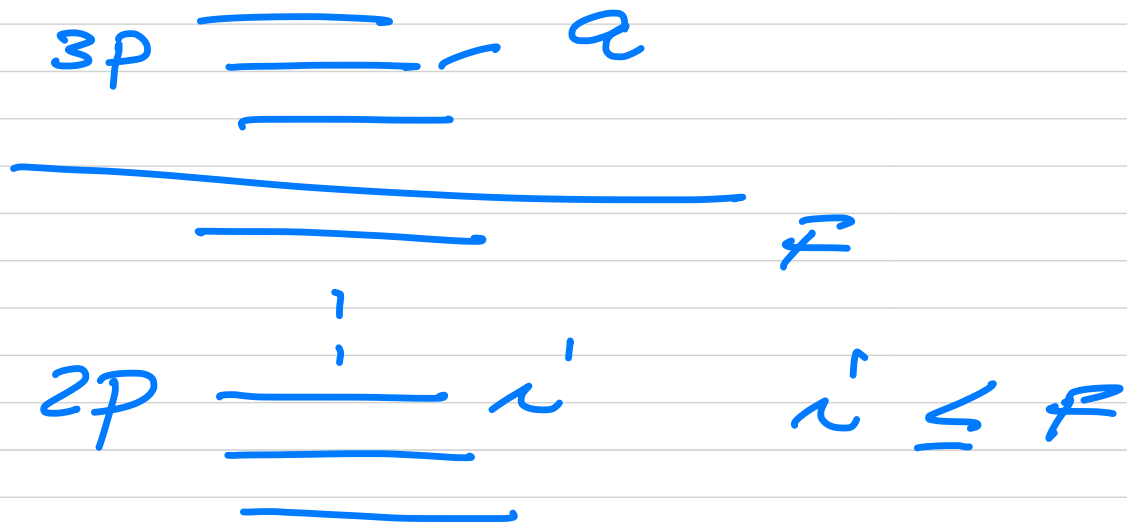
- Then repeat

$$H^{HF}(1) \vec{C}^{(2)} = E^{HF}(2) \vec{C}^{(2)}$$

til convergence criterion
has been reached

$$|E^{HF}(n+1) - E^{HF}(n)| \leq \lambda$$

$$\lambda \sim 10^{-8} \text{ or smaller}$$



$$\psi_i(x) \rightarrow N_{a'} (\psi_i(x) + \delta \psi_i(x))$$

$\uparrow$
 normalization
 constant

$$\delta \psi_i(x) = \eta \psi_a(x)$$

$$|\eta| \ll 1$$

$$N_i^{-2} = \int dx (\psi_i^* + \eta^* \phi_a^*) (\psi_i + \eta \phi_a)$$

$$\langle \phi_a | \psi_i \rangle = 0$$

$$= 1 + |\eta|^2 \quad |\eta| \ll 1$$

$$N_i' \approx 1 \quad \text{at linear order in } \eta$$

$$\Phi_0 \rightarrow \Phi_0^{\text{HF}} + \eta \delta \Phi_0^{\text{HF}}$$

$$E_0^{\text{HF}} = \underbrace{\langle \Phi_0^{\text{HF}} | \hat{H} | \Phi_0^{\text{HF}} \rangle}_{E_{\text{Ref}}^{\text{HF}}} +$$

$$+ [\eta^* \langle \delta \Phi_0^{HF} | H | \Phi_0^{HF} \rangle$$

$$+ \eta \langle \Phi_0^{HF} | H | \delta \Phi_0^{HF} \rangle]$$

$$+ |\eta|^2 \langle \delta \Phi_0^{HF} | H | \delta \Phi_0^{HF} \rangle$$

leave out $|\eta|^2$ $|\eta| \ll 1$

$$\delta E_0^{HF} = 0$$

$$\eta^* \langle \delta \Phi_0^{HF} | H | \Phi_0^{HF} \rangle$$

$$+ \eta \langle \Phi_0^{HF} | H | \delta \Phi_0^{HF} \rangle$$

$$\eta \langle \Phi_0^{HF} | H | \underbrace{\delta \Phi_0^{HF}}_{1p1h} \rangle = 0$$

$$\delta \Phi_0^{HF} = ?$$

$$\delta \Phi_0^{HF} = \frac{1}{\sqrt{n!}}$$

replaces

$$\varphi_i(x_j)$$

$$\begin{array}{c} \varphi_1(x_1) \quad \dots \quad \varphi_1(x_n) \\ \varphi_2(x_1) \quad \quad \quad \varphi_2(x_n) \\ \vdots \quad \quad \quad \vdots \\ \boxed{\varphi_a(x_1) \quad \quad \quad \varphi_a(x_n)} \\ \vdots \quad \quad \quad \vdots \\ \varphi_n(x_1) \quad \quad \quad \varphi_n(x_n) \end{array}$$

$$\langle \Phi_c^{\text{HF}} | \hat{H} | \Phi_c^{\text{HF}} \rangle =$$

$$\langle \hat{c} | \hat{f} | a \rangle = 0$$

$$\langle p | \hat{f} | q \rangle = \langle p | \hat{h}_0 | q \rangle$$

$$+ \sum_{j \leq F} \langle p_j | \hat{v} | q_j \rangle_{\text{AF}}$$

$\hat{U}^{\text{HF}}$ effective mean field.

2nd quantization

$$|\Phi_0^{HF}\rangle = \prod_{i=1}^N a_i^\dagger |0\rangle$$

Define the variation as

$$\eta |\delta \Phi_0^{HF}\rangle = \eta a_a^\dagger a_i |\Phi_0^{HF}\rangle$$

$$\langle \delta \Phi_0^{HF} | \hat{H} | \Phi_0^{HF} \rangle = 0$$

or

$$\langle \Phi_0^{HF} | \hat{H} | \delta \Phi_0^{HF} \rangle = 0$$

$$\hat{H} = \underbrace{E_{\text{ref}}^{\text{HF}}}_{\text{reference energy}} + \hat{F}_N + \hat{V}_N$$

$$\langle \Phi_0^{\text{HF}} | \hat{H} | \Phi_0^{\text{HF}} \rangle$$

$$\langle \sum_p q_p \Phi_p^{\text{HF}} | \hat{H} | \Phi_0^{\text{HF}} \rangle$$

$$= \sum_{p,q} \langle p | q | r \rangle \langle \Phi_0^{\text{HF}} | \overbrace{a_i^\dagger a_a a_p^\dagger a_q}^{\delta_{ap} \delta_{iq}} | \Phi_0^{\text{HF}} \rangle$$

$$= \langle a | f | i \rangle$$

$$= 0$$

standard HF

$$\langle p | \hat{f} | q \rangle = \langle p | \hat{h}^{\text{HF}} | q \rangle$$

$$= \epsilon_p^{\text{HF}} \delta_{pq}$$

$$i \leq F: \langle i | \hat{f} | i \rangle = \langle i | \hat{h}^{\text{HF}} | i \rangle = \epsilon_i^{\text{HF}}$$

$$a > F: \langle a | \hat{f} | a \rangle = \epsilon_a^{\text{HF}}$$

$$\text{and } \langle a | \hat{f} | i \rangle = 0$$

$$\hat{h}^{\text{HF}} | p \rangle = \epsilon_p^{\text{HF}} | p \rangle$$

$$\langle \alpha \beta | \psi | \alpha \beta \rangle_{AS} \longrightarrow \langle p q | \psi | p q \rangle_{AS}$$

FCI from last week

	0p0h	1p1h	2p2h	3p3h	-	-
0p0h	$E_0^{\text{Ref}}(\Phi_0)$	$\langle \alpha f \alpha \rangle_{SF}$	$\times$	0	0	-
1p1h	x	x				
2p2h	x	x				
3p3h	0	x				
1	1	0				
1	1					

HF and FCI

	0pck	1p1k	2p2k	3p3k	...
0pck	E_{ref}^{HF}	0	$\times 2$	0	0
1p1k	0	$\times 2$	$\times 2$	$\times 2$	0
2p2k	$\times 2$	$\times 2$	$\times 2$	$\times 2$	
3p3k	0	$\times 2$	$\times 2$	$\times 2$	
1	1	0			
1	1				
1					
.					

$$\underline{S}^+ H \underline{S} = D$$

$$\underline{S}^+ \underline{S} = \underline{S} \underline{S}^T = \underline{1}$$

Thouless' theorem

a general SD can be written as

$$|C'\rangle = \exp\left(\sum_{a_i} c_i a_i^\dagger a_i\right) |C\rangle$$

$$|\Phi_0^{HF}\rangle$$

CIS (config int with
singles) $\propto |\Phi_0\rangle$

$$|\psi_0\rangle \simeq |\Phi_0\rangle + \sum_{a_i} c_i a_i^\dagger a_i |\Phi_0\rangle$$

T_1 (coupled-cluster)

$$T_1 = \sum t_a^a q_a^\dagger q_a$$

↑ unknown amplitudes
Proof of Theorem 1 theorem

$$(i) \exp(\hat{q} + \hat{r}) = \exp(\hat{q}) \exp(\hat{r})$$

$$(ii) e^{\sum_{i=1}^N x_i} = e^{x_1} e^{x_2} \dots e^{x_N}$$

$$= \prod_{i=1}^N e^{x_i}$$

$$(iii) \sum_{q \in F} = \sum_{i \in F} \sum_{q \in F}$$

$$|C'\rangle = \prod_{i' \in F} \left(1 + \sum_{a \in F} c_{i'}^a a_a^\dagger a_{i'} \right)$$

$$+ \frac{1}{2} \left(\sum_{a \in F} c_{i'}^a a_a^\dagger a_{i'} \right)^2 + \dots \Big) |C\rangle$$

$$\left(\sum_{a \in F} c_{i'}^a a_a^\dagger a_{i'} \sum_{a' \in F} c_{i'}^{a'} a_{a'}^\dagger a_{i'} \right) |C\rangle$$

$$i' \in C$$

$$= \prod_{i' \in F} \left(1 + \sum_{a \in F} c_{i'}^a a_a^\dagger a_{i'} \right) |C\rangle$$